

article amsmath enumerate
 article amsmath enumerate Chapter 2: Thermodynamic Properties and
 Molecular Dynamics

2.1 Pressure from Molecular Collisions

Molecular Collision Dynamics

- **$D =$** *This equation represents the relationship between molecular collisions and the dynamic process of gas molecules in*

- **3. Momentum Transferred to the Wall** - **$h_j = 2i E_j$** : The momentum transferred per collision is given by this formula, where '2i' refers to mass and 'E_j' represents velocity in a specific direction.

- **4. Total Transferred Momentum** - **$j = i \cdot 50 = 0.50$** - **$j = 5.0$** : This equation describes the total pressure generated by molecular collisions, where 'i', 'j', and 'o' are mass-related constants and variables.

2.2 The Ideal Gas Assumption

- **Ideal gas**: An ideal gas is a theoretical model of gases in which molecules have no intermolecular interactions or attraction/repulsion forces.

- **Elastic Collisions**: - Elastic collisions between molecules and the wall are not necessary if the system is at thermal equilibrium, meaning '8.666668'.

2.3 Equipartition of Energy

The energy distribution in an ideal gas can be analyzed using statistical mechanics:

1. **Translational Kinetic Energy**: - Each degree of freedom contributes 4.0 to the average translational kinetic energy.

2. **Rotational Degrees of Freedom**: - Rotational degrees are excited if atoms' centers of gravity are not aligned with the rotation axis.

3. **Vibrational Degrees of Freedom**: - Vibrational energy per degree of freedom is 4.0, where quanta of energy " must be considered for quantum effects at low temperatures.

Summary Table: Degrees of Freedom

Number of Atoms	yK	yL	yM	Example
1	3	0	0	Ar
2	3	2	1	N ₂
3	linear	3	2	O=C=O
3	non-linear	3	3	H ₂ O
<i>N, linear</i> $ 3 2 3_N-5 H-CC-H _{N,n on-linear} 3 3 3_N-6 C_2H_5 $				

2.4 Thermodynamic Properties and Degrees of Freedom

- **Degrees of freedom** (s

- **Energy per molecule**: - Translation: 4.0 - Rotation: 4.0 - Vibration: 2.4.0

Remarks

1. **Boltzmann's Constant**: Understanding the relationship between Boltzmann's constant and the universal gas constant.

2. **Phase Transition Dynamics**: - The behavior of isentropic expansion in the two-phase area liquid-vapor can have a positive inclination if conditions meet certain criteria.

Key Equations

- **Momentum Transfer**: $h_j = 2i E_j$

- **Total Momentum and Pressure**: - $j = i \cdot 50$ - $j = 5.0$

- **Energy Contribution from Degrees of Freedom**: - $r = (s$

Examples for Calculation at 300K - Specific examples and calculations would be included here, considering the energy contributions based on degrees of freedom.

****Notes:**** - Understanding the interplay between molecular dynamics (collisions, rotation, vibration) and thermodynamic properties (pressure, temperature) is crucial. - The role of quantum mechanics in determining vibrational excitation at low temperatures needs careful consideration.